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CO4-AT2 Industry Problem Solving

**COURSE NAME: FUNDAMENTALS OF
DATA SCIENCE**

COURSE CODE: DSA0404

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Que: A bank wants to estimate the average monthly transaction value of its customers. A sample of 200 customers has a mean transaction value of ₹18,500 and standard deviation of ₹4,200. Calculate the point estimates of the population mean and variance.

Problem Statement

A bank wants to estimate the **average monthly transaction value** of its customers. A random sample of **200 customers** gives:

- Sample size:
- Sample mean:
- Sample standard deviation:

Calculate the **point estimates of the population mean and population variance**.

1. Introduction to Point Estimation

Point estimation is a statistical method used to estimate an unknown population parameter using a single numerical value calculated from a sample.

In this banking problem, the bank is interested in two population parameters:

1. **Population mean** – the true average monthly transaction value of all customers.
2. **Population variance** – the true variability in monthly transaction values of all customers.

Since the bank cannot collect transaction information from every customer, it takes a random sample of 200 customers and uses the sample statistics as estimates.

Basic idea

For example: $\bar{x} \rightarrow \mu$ and $s^2 \rightarrow \sigma^2$

2. Point Estimate of Population Mean

The **sample mean** is the point estimator of the population mean.

The formula is:

$$\hat{\mu} = \bar{x}$$

Given: $\bar{x} = ₹18,500$

Therefore, $\mu^{\wedge} = ₹18,500$

Interpretation

The **point estimate of the population mean monthly transaction value is ₹18,500.**

This means that, based on the sample of 200 customers, the bank estimates that the average monthly transaction value of its entire customer population is approximately **₹18,500 per customer.**

3. Point Estimate of Population Variance

The sample standard deviation is: $s = ₹4,200$

The population variance is estimated using the **sample variance:**

$$\hat{\sigma}^2 = s^2$$

Therefore,

$$s^2 = (4,200)^2$$

$$s^2 = 17,640,000$$

Thus,

$$\hat{\sigma}^2 = ₹17,640,000^2$$

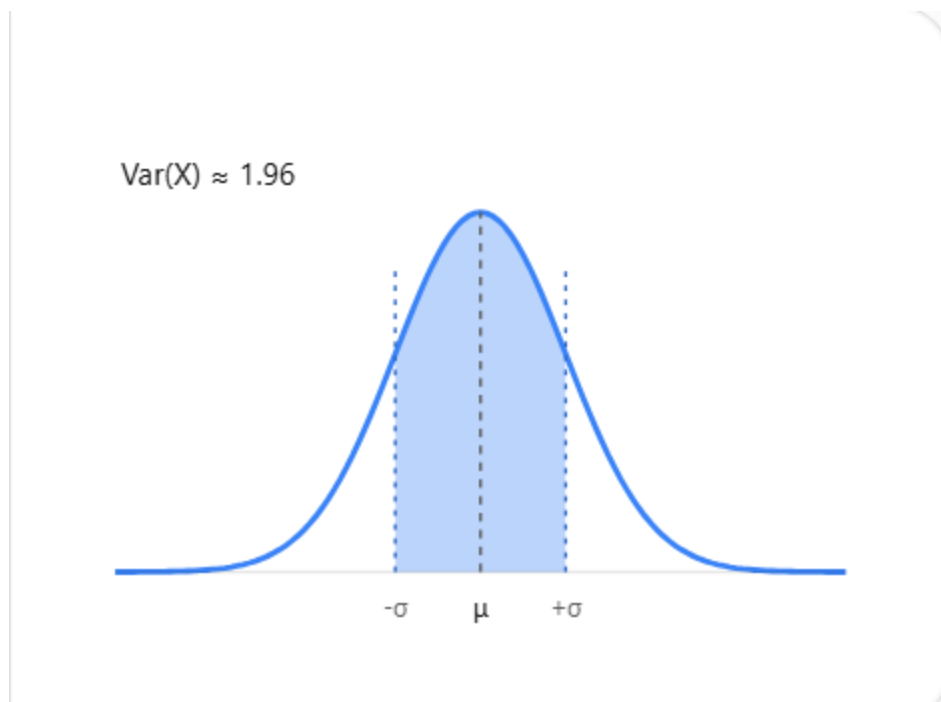
More precisely, since variance is measured in **squared rupees:**

$$\hat{\sigma}^2 = 17,640,000 \text{ rupees}^2$$

A larger variance indicates greater variation in the transaction values among customers.

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{Var}(X) = (1.4)^2 = 1.96$$



4. Why Are Sample Statistics Used?

The bank wants information about **all customers**, but collecting the monthly transaction value of every customer may be expensive and time-consuming. Therefore, a random sample of 200 customers is selected.

The sample provides:

$\bar{x} = ₹18,500$ and $s = ₹4,200$

5. Important Formulae

Population Mean

$$\hat{\mu} = \bar{x}$$

Population Variance

$$\hat{\sigma}^2 = s^2$$

If the sample standard deviation is given:

$$s^2 = (s)^2$$

For this problem:

$$s^2 = (₹4,200)^2 = 17,640,000 \text{ rupees}^2$$

6. Step-by-Step Numerical Calculation

Step 1: Identify the sample information

$$n=200$$

$$\bar{x}=\text{₹}18,500$$

$$s=\text{₹}4,200$$

Step 2: Estimate population mean

$$\mu^{\wedge}=\bar{x}$$

$$\mu^{\wedge}=\text{₹}18,500$$

Step 3: Estimate population variance

$$\sigma^2=s^2$$

$$\sigma^2=(\text{₹}4,200)^2$$

$$\sigma^2=\text{₹}17,640,000 \text{ rupees}^2$$

7. Banking Industry Interpretation

The result can be interpreted from a banking perspective.

The bank estimates that its customers have an **average monthly transaction value of ₹18,500**.

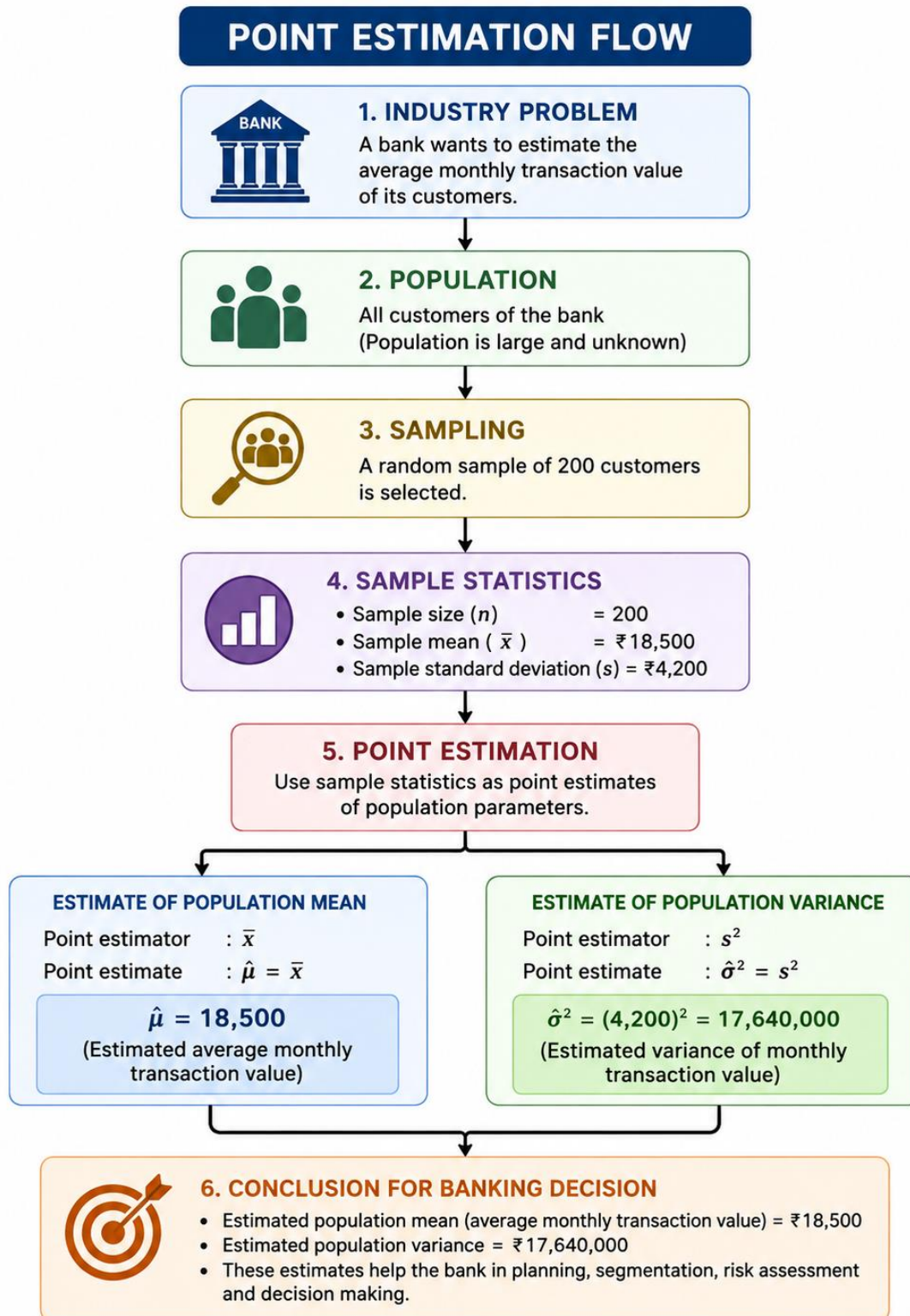
The estimated variance is: 17,640,000 rupees²

The corresponding sample standard deviation of **₹4,200** indicates that individual customers' monthly transaction values can vary considerably around the average.

This information can help the bank in:

- customer segmentation,
- transaction-volume analysis,
- financial planning,
- credit and risk assessment,
- designing banking products,
- identifying high-value customers, and
- estimating expected transaction volumes.

8. Point Estimation Flow



Final Answer

Parameter	Formula	Calculation	Point Estimate
Population Mean μ	$\mu^{\wedge} = \bar{x}$	₹18,500	₹18,500
Population Variance	$\hat{\sigma}^2 = s^2$	(4,200 ²)	17,640,000 ₹ ²

Conclusion

Therefore, using the sample of **200 customers**, the bank's point estimates are:
Hence, the bank can use **₹18,500 as the estimated average monthly transaction value** for its entire customer population, while **17,640,000 ₹²** represents the estimated variability of transaction values.